



Experimental and kinetic modeling study of laminar burning velocities of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames

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ABSTRACT

Ammonia (NH_3) is a promising energy carrier to store and transport renewable hydrogen (H_2) that can be generated using, e.g., wind and solar energy. Direct combustion of NH_3 is one of the possible methods to utilize the energy by the end users. To understand the combustion characteristics of NH_3 as a fuel, the laminar burning velocities of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames were investigated experimentally using the heat flux method. Measurements are reported for a wide range of equivalence ratios and blending ratios. Kinetic modeling was also performed using available chemical kinetic mechanisms, namely the GRI-Mech 3.0, the Okafor et al. and the San Diego mechanisms. The experimental results for NH_3/air flames agree well with the literature data and it is found that blending NH_3 with H_2 is the most effective manner to increase the burning velocity of NH_3 based fuel mixtures. None of the kinetic mechanisms used can accurately predict most of the measured data. Sensitivity and reaction path analyses indicate that the oxidation of NH_3 blended with the additive fuels considered can be understood as the parallel oxidation processes of the individual fuels, and that the source of discrepancy between the experimental and modeling results is related to the inaccuracy of the rate parameters of the N-containing reactions. In this regard, the present detailed and reliable experimental data is of special value for model development and validation.

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1. Introduction

Nowadays, it is highly required to reduce CO_2 emissions from human activities to mitigate the increasing global warming [1]. One of the key methods to meet the CO_2 emission targets is to increase the utilization of renewable, low- or non-carbon fuels in power and energy systems. Besides the well-recognized hydrogen (H_2), ammonia (NH_3) has also been proposed as an energy carrier to store and transport intermittent renewable energy sources, like solar and wind, over a long distance with a much lower cost compared with liquefied H_2 [2–5]. Direct combustion of NH_3 as a fuel may be used efficiently in, e.g., engines, boilers and gas turbines [6–8]. However, it has already been recognized that NH_3 requires high energy for attainment of ignition and presents low burning velocities compared with conventional hydrocarbon fuels [9,10]. To overcome these disadvantages, it is necessary to establish effective methods to burn NH_3 . Co-firing NH_3 with fuels of higher reactivity, e.g., H_2 , has been proposed [6–8]. However, information avail-

able on key combustion features, such as laminar burning velocity (S_L), is still insufficient and often inconsistent among the limited studies. This is true not only for NH_3 blended with H_2 but also for blends of NH_3 with other fuels, e.g., carbon monoxide (CO) and methane (CH_4), and even for pure NH_3 . This gap needs to be filled to develop reliable chemical kinetic mechanisms for NH_3 combustion, which is critical for industrial utilization of this carbon-free fuel.

Related previous studies on laminar burning velocities of NH_3 and its blends with other fuels in air are rather scarce. Lee et al. [11] measured the laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures as a function of the mole fraction of H_2 (x_{H_2}) in the fuel mixture. The measurements were performed at room temperature and atmospheric pressure using the outwardly propagating spherical flame method. Ichikawa et al. [12] used the same method to measure the laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures, also as a function of x_{H_2} , but at elevated pressures up to 0.5 MPa. Using Bunsen flames at atmospheric pressure, Kumar and Meyer [13] measured laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures also as a function of x_{H_2} . A similar work was reported by Li et al. [14]. These studies have significantly increased the understanding on the enhancement effects of H_2 on NH_3 combustion, but the

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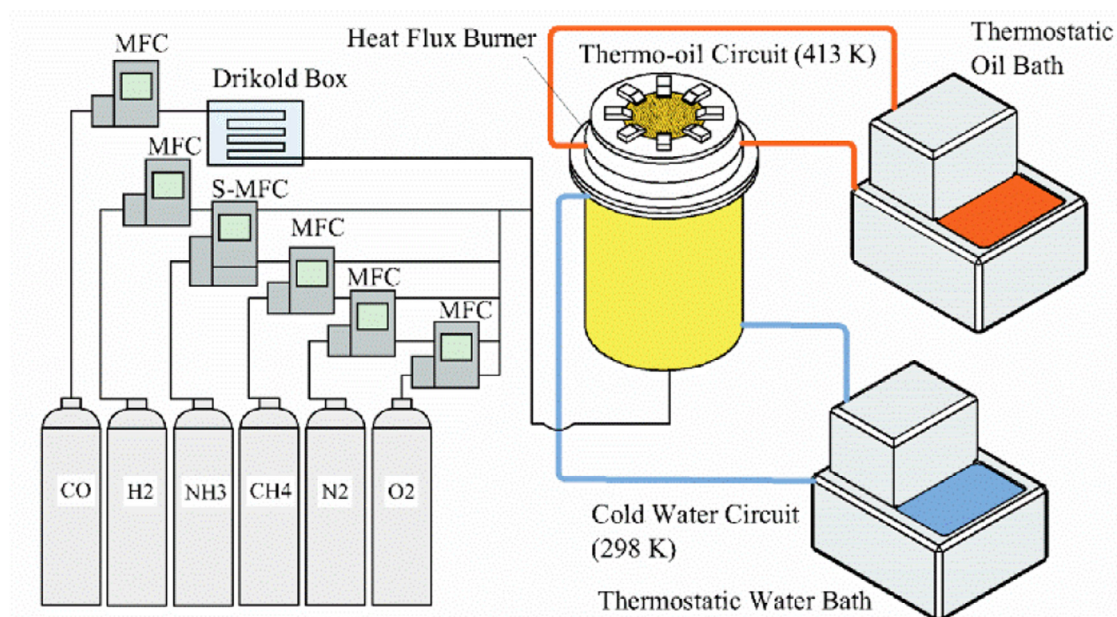


Fig. 1. Schematic diagram of the experimental setup.

data available are still limited considering the complexity of the subject.

Similar studies of NH_3 blending with other fuels, such as CH_4 and syngas, are even rarer. However, it should be noted that co-firing NH_3 with natural gas or syngas is also promising and thus S_L data for $\text{NH}_3/\text{CH}_4/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ mixtures are also needed. To the best of our knowledge, laminar burning velocities of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures as a function of the mole fraction of CH_4 (x_{CH_4}) in the fuel mixture were reported only by Okafor et al. [15]. In addition, reference should be made to studies where small amount of NH_3 were added to CH_4 flames to study the nitrogen chemistry related to NO formation and emission, e.g., [16–18].

In the limited experiments mentioned above, investigators usually used spherical or Bunsen flames. Spherical flames are influenced by the stretch effect on the flame front, while Bunsen flames are influenced by both stretch effect and heat losses. The usage of extrapolation methods to obtain stretchless flames may introduce uncertainties in the measurement of S_L . Alternatively, the heat flux method [19–21] is one of the most attractive methods for measuring S_L . This method is based on an adiabatic stretchless flat flame [22] and has important advantages over other methods, such as good data consistency and experimental uncertainties as low as $\pm 1 \text{ cm/s}$ [23,24]. Such a low measurement uncertainty is necessary in studying a fuel with relatively low burning velocities, such as NH_3 . However, the heat flux method has not been demonstrated for measuring NH_3 flames. Thus, further assessment of the technique may also be required.

As for kinetic modeling, several NH_3 containing mechanisms used to predict S_L for NH_3 based flames showed large discrepancies between the measured and predicted data [14,15]. The discrepancies may be because that in these NH_3 containing mechanisms, e.g., the Konnov mechanism [25], NH_3 is treated as an intermediate species rather than a main reactant and some important reactions related to NH_3 combustion may have been neglected. Therefore, it is also necessary to evaluate the applicability of the existing NH_3 containing mechanisms to NH_3 flames, especially for blends of NH_3 with different fuels, e.g., H_2 , CO , and CH_4 .

In light of the background above, the specific aims of this investigation are (1) to provide accurate laminar burning velocities for NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames

as a function of the equivalence ratio and of the mole fraction of the components in the fuel mixture; and (2) to assess the performance of three chemical kinetic mechanisms and investigate the reasons behind the large discrepancies between experimental and predicted data by kinetic analyses.

2. Experiment

2.1. Heat flux measurement system

Figure 1 presents a diagram of the experiment setup used in this work, which has been described in detail elsewhere [26–28]. Briefly, it comprises of a heat flux burner and a gas feeding system. Compared with our previous setup, two modifications were made in the setup for the present work. First, a corrosion-resistant mass flow controller (MFC) was used for the corrosive NH_3 stream. Second, thermo-oil instead of water was used in the fluid circuit around the burner to increase the burner plate temperature (T_p). When T_p was set at 363 K, which was used in our previous experiments for hydrocarbon flames, the edge of the NH_3 flame lifted up significantly, as shown in Fig. 2a. This phenomenon is attributed to the cooling effect of the surrounding air on the flame, which is particularly significant for a fuel, like ammonia, with both high ignition energy and low burning velocity. To mitigate this tilting on the flame edge, T_p was increased to 413 K by using thermo-oil and thereby attaining a flatter flame, as shown in Fig. 2b. The gaseous mixture at the inlet absorbs more heat from the burner plate, which enhances the combustion of NH_3 . It is worth noting that the heat flux method is nearly independent of T_p [29] since an increase in T_p does not influence the heat flux balance used to

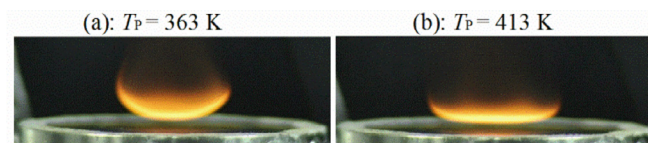


Fig. 2. Photographs of a stoichiometric NH_3/air flame with different burner plate temperatures: (a) $T_p = 363 \text{ K}$ and (b) $T_p = 413 \text{ K}$.

measure S_L [22]. This also indicates that increasing T_p may be very beneficial in investigations of other hard-to-ignite fuels.

Figure 2(b) discloses that there was still a small tilting on the flame edge, which slowly fluctuated up and down. We concluded that this small tilting has a negligible influence on the measured S_L for the following reason. In this heat flux method, the temperature profile of the burner surface plate was measured by eight thermocouples at different radial distances and was used to determine the inlet flow velocity that equals the laminar burning velocity. The measured temperatures have small intrinsic fluctuations due to the use of thermocouples. In our experiments, it was found that for the flat CH_4/air flames with relatively high burning velocities, the fluctuations of the measured temperature profile result only from this intrinsic fluctuation, while for the non-flat NH_3 flames (cf. Fig. 2b), the fluctuations of the measured temperature result from the combined effects of the intrinsic fluctuations of the thermocouples and the edge tilting. Our experiments revealed that the fluctuations of the temperatures measured are similar for the CH_4/air and NH_3/air flames. This indicates that the influence of the non-flatness of the NH_3 flame edge on the measured S_L is significantly less important than the intrinsic fluctuation and is a negligible effect when T_p equals 413 K.

All gases used in this work were purchased from the Jingong Gas (Hangzhou, China) with purities higher than 99.99%. The gases were fed into the burner through stainless steel pipes, except for CO, which was fed through a Teflon pipe to avoid the reaction between iron and CO to form iron carbonyl ($\text{Fe}(\text{CO})_5$). $\text{Fe}(\text{CO})_5$ has been reported as a strong flame inhibitor [30]; indeed, even a small amount of $\text{Fe}(\text{CO})_5$ can significantly lower the burning velocity, e.g., of rich syngas flames [31,32]. In addition, the CO cylinder pressure regulator may also contribute to produce metal-carbonyl compounds so that a cold trap made of drikold, $\sim 195\text{ K}$, was incorporated into the plastic pipe to capture trace contaminants. All unburnt mixtures in this study were made at 298 K and atmospheric pressure.

2.2. Buoyancy influence on the S_L measurements

Since the burning velocity of NH_3/air flames is very low, the influence of buoyancy, or Rayleigh–Taylor instability, on the measurements of laminar burning velocities has been mentioned by several researchers who used the spherical flame method, e.g., Ichikawa et al. [12] and Lee et al. [11]. In the work of Lee et al. [11], buoyant instability was detected by “distortion of the flame surface from a spherical shape” and by “upward motion of the centroid of the flame image”. In contrast, the flame stabilized on a heat flux burner is always downwardly propagating. Because the denser, unburned mixture is always below the lighter, burned mixture, the flames investigated in the present experiment should be buoyantly stable [33]. The inspection of flames confirmed that the observations reported for spherical flames do not apply to the present experiments.

2.3. NH_3 adsorption influence on the S_L measurements

It is well-known that NH_3 tends to adsorb on the surface, which originates that (a) the actual NH_3 fed into the flame may deviate from that established in the upstream MFC; and (b) the adsorbed NH_3 may react with the brass metal of the burner chamber and surface, leading to changes in the downstream flame structure.

During the present experiments, color changes on the burner plate surface was observed after a few days of experiments with NH_3 , indicating possible NH_3 adsorption on the burner surface. The color changes observed in the present work was identical to that described by Dyakov et al. [34], who concluded that the measured

burning velocities are not affected by this phenomenon. The reaction between NH_3 and the heat flux metal surface produces colored deposits, e.g., [35,36]. In the present experiments, no colored deposits were perceptible so that this phenomenon should be negligible. Thus we conclude that the adsorption of NH_3 should have negligible influence in the present experiments.

2.4. Initial flow conditions and experimental uncertainties

Each studied case is here referred to by the equivalence ratio ϕ and by the mole fraction x of the components in the fuel mixture. The equivalence ratio is defined as $\phi = \frac{F/A}{(F/A)_s}$ [37] and the mole fraction of M ($= \text{H}_2$, CO , or CH_4) in the fuel mixture as $x_M = V_M/V_{\text{fuel mixture}}$. The initial conditions for all studied cases are listed in the Supplementary Material.

There are two main uncertainty sources in the present measurements of S_L . One is the uncertainty of the burner plate temperatures measured using the eight thermocouples, which is defined as the ratio of the measured temperature scatter (σ_c) to the measurement sensitivity (s) [38]. The other is the uncertainty of the gas flow rates, which are controlled using Alicat Scientific MFCs. According to the specifications provided by the supplier, the present mass flow controllers have a standard accuracy calibration of $\pm (0.8\% \text{ of reading} + 0.2\% \text{ of full scale})$. For different initial flow rates of the flames, uncertainties were calculated and presented with the S_L results. More details on experimental uncertainties can be found in our previous study [39] and in the review of Alekseev et al. [38].

3. Kinetic modeling

Three chemical mechanisms were used in the present work, namely the GRI-mech 3.0 [40], the mechanism of Okafor et al. [15] and the mechanism of San Diego [41]. The GRI-mech 3.0 mechanism was developed for CH_4 combustion and has been widely validated. The mechanism of Okafor et al. [15] was recently released and has been applied to calculate S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ flames. This mechanism is based on the GRI-mech 3.0 and includes some important reactions extracted from the Tian et al. mechanism [16]. The San Diego mechanism is a combination of the nitrogen chemistry sub-mechanism, updated on 2018/07/23, and the main mechanism containing C species, updated on 2016/12/14 [41]. The nitrogen chemistry sub-mechanism has been targeted for NH_3 ignition processes [41]. In addition to the three mechanisms, three other chemical mechanisms were also applied for kinetic analysis of some of the cases studied, of which the details are presented in Sections 4.1.4 and 4.3.

All the kinetic modeling in the present study was performed using the CHEMKIN-III [42] using the PREMIXED module. For each case studied, at least 500 grid points were used with values of GRAD and CURV setting at 0.02.

4. Results and discussion

4.1. Experimental results

4.1.1. CH_4/air flames

Initially, measurements of S_L were performed for CH_4/air and NH_3/air flames, and the results were compared with those in the literature to validate the reliability of the present experimental method.

Figure 3 shows laminar burning velocities of CH_4/air flames, as a function of ϕ , measured in the present work and those in the works of Bosschaart and De Goey [19], Dirrenberger et al. [43], Hu et al. [44], Wu et al. [45] and Okafor et al. [15]. The data of Bosschaart and De Goey [19] and Dirrenberger et al. [43] were also

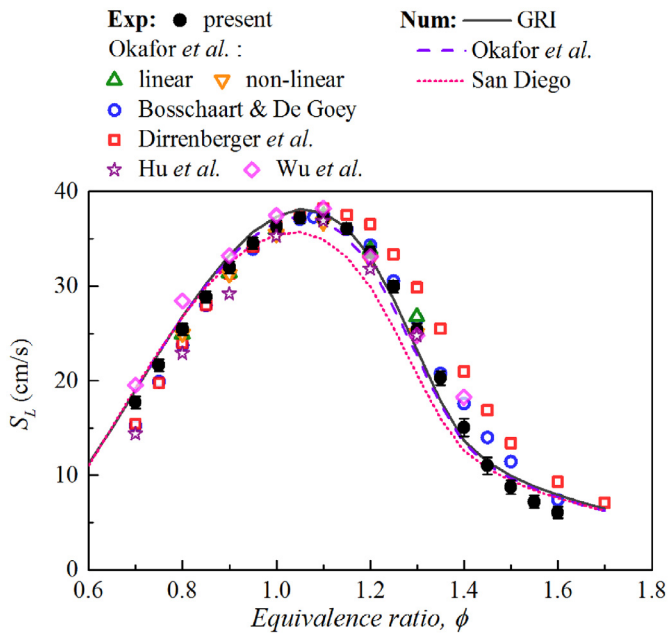


Fig. 3. Laminar burning velocities of CH_4/air flames, as a function of ϕ , measured in this work and those reported in the works of Bosschaart and De Goeij [19], Dirrenberger et al. [43], Hu et al. [44], Wu et al. [45] and Okafor et al. [15], and predicted laminar burning velocities using three mechanisms.

obtained using the heat flux method, while the data of Hu et al. [44] and Okafor et al. [15] were obtained using the spherical flame method. Only the data of Wu et al. [45] were obtained using the Bunsen flame method. The present results agree well with the literature data for $0.7 \leq \phi \leq 1.3$, but are slightly lower for $\phi > 1.3$. The general good agreement between the present results and those in the literature indicates that the present arrangement works well for CH_4/air flames.

Figure 3 also shows the laminar burning velocities predicted using the three chemical mechanisms. It can be seen that both the GRI-mech and the Okafor et al. mechanism reproduce very well the experimental data, regardless the value of ϕ , while the San Diego mechanism predicts slightly lower values of S_L for ϕ between 0.7 and 1.3. The overall good agreement between the predictions and measurements indicates that the three mechanisms work well for CH_4/air flames.

4.1.2. NH_3/air flames

Figure 4 shows laminar burning velocities of NH_3/air flames, as a function of ϕ , measured in the present work and in previous works of Takizawa et al. [10], Ronney [46], Pfahl et al. [47], Jabbour and Clodic [48], Hayakawa et al. [49] and Ichikawa et al. [12]. With the exception of Jabbour et al. [48], who obtained the data using a vertical tube, all the other investigators obtained the data using the spherical flame method. Our S_L data present a similar evolution with the equivalence ratio as that reported in the literature, but the present values are always lower by approximately 1 cm/s. It is important to point out that the experimental data of Ronney [46], Pfahl et al. [47], Jabbour et al. [48] and Takizawa et al. [10] were not corrected for stretch effects, which could reduce the accuracy of their results, as discussed by Lee et al. [11]. Albeit the discrepancy between the data, Fig. 4 indicates that the present setup is reliable to measure the low S_L values of NH_3/air flames with high accuracy. The maximum S_L uncertainty in Fig. 4 presents an absolute value of ± 1.0 cm/s or a relative value of $\pm 33.3\%$ at $\phi = 0.85$. This estimated uncertainty is the largest value for all conditions studied in this work.

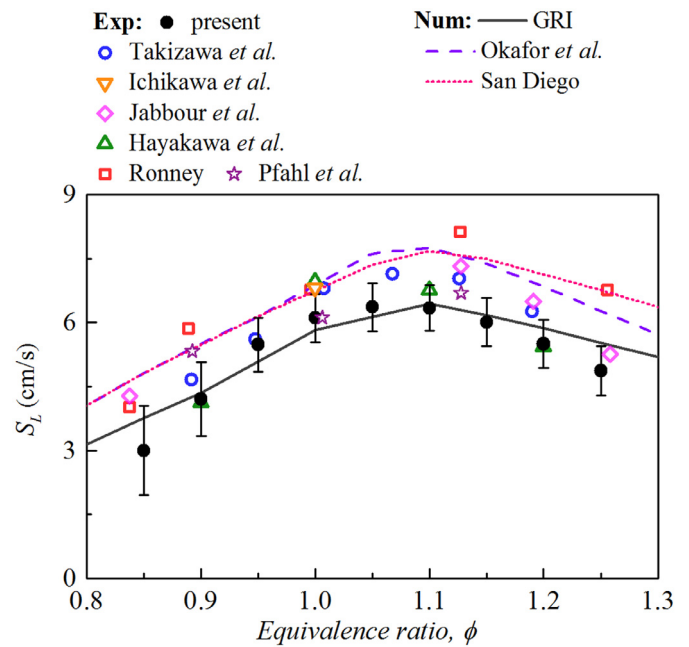


Fig. 4. Laminar burning velocities of NH_3/air flames, as a function of ϕ , measured in this work and in the works of Takizawa et al. [10], Ronney [46], Pfahl et al. [47], Jabbour et al. [48], Hayakawa et al. [49] and Ichikawa et al. [12], and predicted laminar burning velocities using three mechanisms.

Figure 4 also shows predicted laminar burning velocities using the three mechanisms considered here. In general, the three mechanisms are able to reproduce reasonably well the experimental data, and it is clear that they work better for CH_4/air flames than for NH_3/air flames.

4.1.3. $\text{NH}_3/\text{H}_2/\text{air}$ flames

Figure 5 shows laminar burning velocities of stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$ flames, as a function of x_{H_2} , measured in the present work and in the works of Kumar and Meyer [13], Li et al. [14], Lee et al. [11] and Ichikawa et al. [12]. Due to the limited range of the mass flow controllers used, measurements could not be made for x_{H_2} larger than 0.45 (cf. Fig. 5b). It can be seen that the S_L value increases with x_{H_2} with a non-linear dependence and the present experimental results are in good agreement with the literature data. The data from this work also show better consistency than the literature results, in particular, better than that reported by Ichikawa et al. [12] and Li et al. [14], who used methods based on spherical or Bunsen flames. Figure 5 also shows the predicted laminar burning velocities using the three mechanisms. All mechanisms under-predict S_L for x_{H_2} ranging from 0 to 0.45, with the San Diego mechanism being the most accurate and the GRI-mech being the most inaccurate.

Figure 6 shows laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$ flames, as a function of ϕ , measured for $x_{\text{H}_2} = 0.15, 0.25, 0.35$ and 0.4. For $x_{\text{H}_2} = 0.15$, no data was measured for lean ($\phi < 0.75$) or rich ($\phi > 1.40$) flames because it was too hard for the flames to be stabilized on the burner surface. It is obvious that the enhancement effect of H_2 on the NH_3 flames applies for all ϕ studied from 0.7 to 1.6 – not only for the stoichiometric condition (Fig. 5). It is also found that the value of ϕ with the maximum S_L slightly increases with x_{H_2} , being 1.05 for $x_{\text{H}_2} = 0$ (cf. Fig. 4), 0.15, 0.25 and 0.35, but 1.10 for $x_{\text{H}_2} = 0.4$. This trend is consistent with the finding that the maximum S_L of H_2/air flames occurs at $\phi = 1.8$ with a value of 286 cm/s [50].

Figure 6 also shows the predicted laminar burning velocities using the three mechanisms. It can be seen that none of

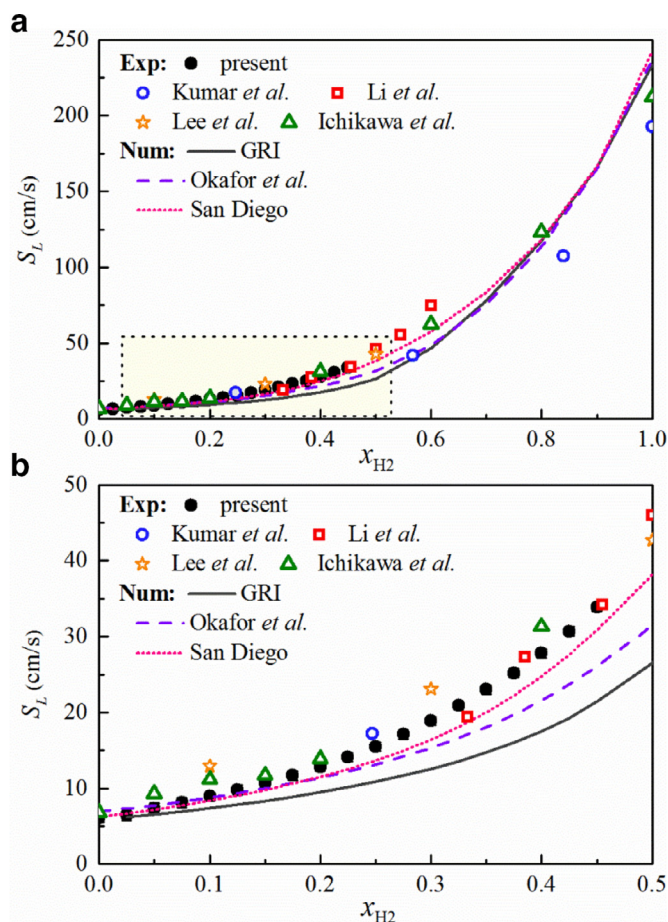


Fig. 5. (a) Laminar burning velocities of stoichiometric NH_3/H_2 /air flames, as a function of x_{H_2} , measured in this work and in the works of Kumar et al. [13], Li et al. [14], Lee et al. [11], Ichikawa et al. [12], and predicted laminar burning velocities using three mechanisms. (b) Enlarged view of the highlighted spot in (a).

the mechanisms can predict the measured data accurately. Large discrepancies exist not only between the experimental and computed results, but also among the computed ones based on the different mechanisms. The San Diego mechanism under-predicts S_L for low values of ϕ , but over-predicts it for high values of ϕ . The Okafor et al. mechanism works relatively well for $x_{H_2} = 0.15$, but in general under-predict S_L for all other conditions. The GRI-mech 3.0 significantly under-predicts S_L for almost all conditions, which demonstrates the relatively weak capability of this widely used mechanism for predicting NH_3 flames.

4.1.4. NH_3/CO /air flames

Figure 7 shows laminar burning velocities of stoichiometric NH_3/CO /air flames, as a function of x_{CO} . No S_L data was reported previously for NH_3/CO /air flames, thus no previous results are included in Fig. 7 for comparison purposes. The three mechanisms can predict reasonably well the experimental trend and the maximum S_L location at $x_{CO} \approx 0.85$. In addition, good agreement can be found between the experimental and predicted data for the mixtures with low and high x_{CO} , while large discrepancies exist for the mixtures with $x_{CO} \approx 0.5$. It is also noted that the GRI-mech presents larger discrepancies than the other two mechanisms.

To assess the reasons for the large discrepancies of the predictions at $x_{CO} \approx 0.5$, three additional literature-available mechanisms were used here, namely, the Tian et al. mechanism [16], the Mendiara and Glarborg [51] mechanism, and the Glarborg et al. mech-

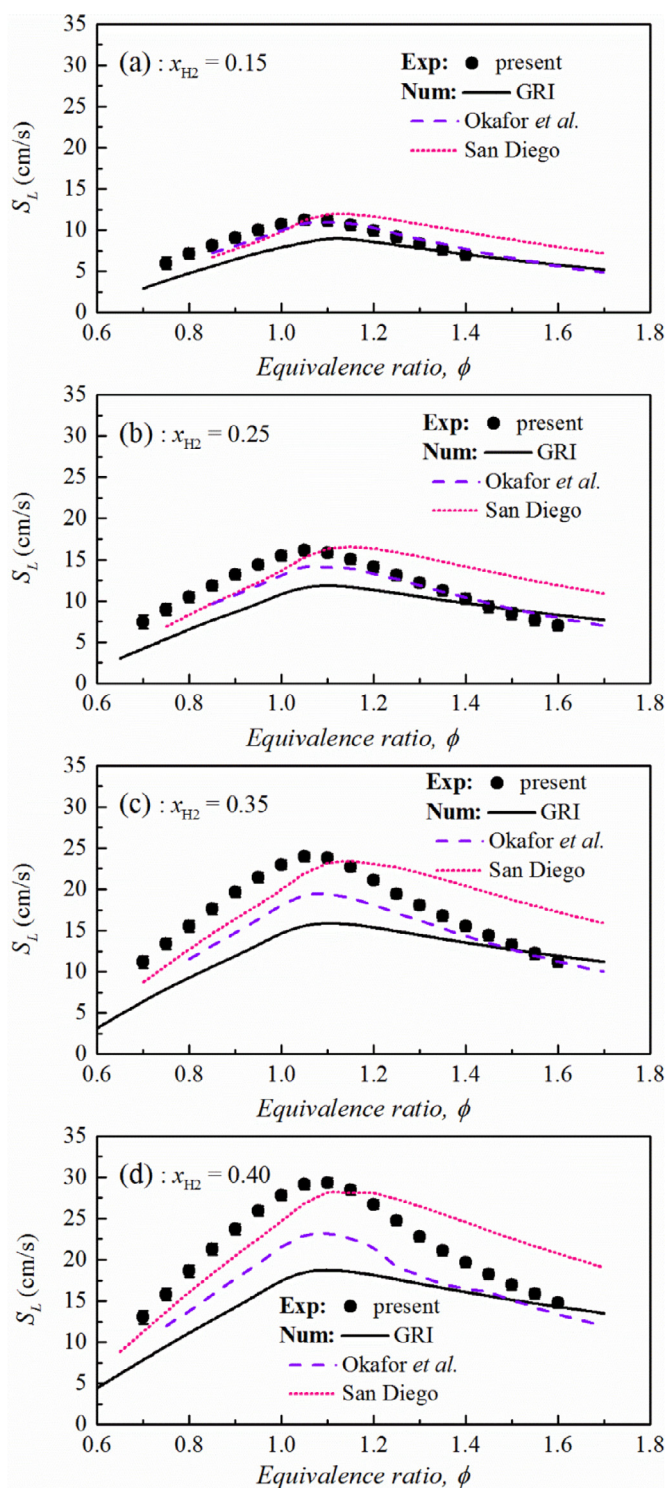


Fig. 6. Laminar burning velocities of NH_3/H_2 /air flames, as a function of ϕ , measured in the present work for $x_{H_2} = 0.15, 0.25, 0.35$ and 0.4 , together with predicted laminar burning velocities using the three mechanisms.

anism [52]. We excluded from this study the use of other existing chemical mechanisms, e.g., the Konnov mechanism [25], because recent modeling studies on NH_3 flames [5, 53] demonstrated that their performance is not better than those used here. Although the mechanisms of Tian et al. [16], Mendiara and Glarborg [51], and Glarborg et al. [52] include more species and reactions related to the NH_3 oxidation and its interaction with C/H/O compounds, Fig. 7 shows that they are still unable to predict the present

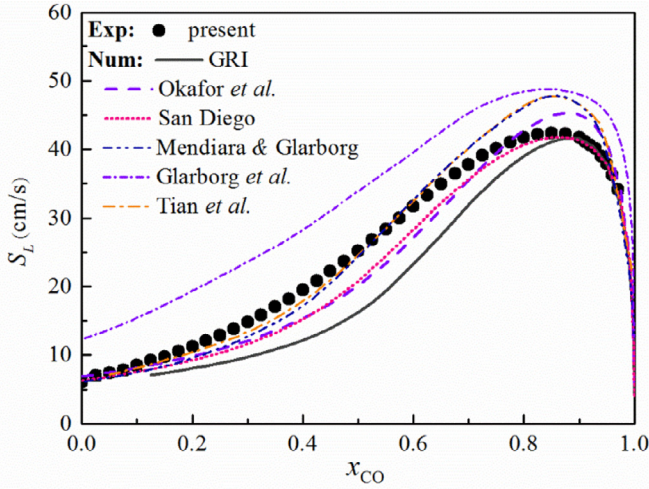


Fig. 7. Laminar burning velocities of stoichiometric $\text{NH}_3/\text{CO}/\text{air}$ flames, as a function of x_{CO} , measured in this work, and predicted laminar burning velocities using different kinetic mechanisms.

experimental data in the whole range of CO mole fraction. The mechanisms of Tian et al. [16] and Mendiara and Glarborg [51] over-predict the data for $x_{\text{CO}} > 0.7$, while the Glarborg et al. mechanism [52], which includes more species and reactions than the mechanisms of Tian et al. [16] and Mendiara and Glarborg [51], over-predicts considerably the experimental data in the whole range. These discrepancies are further discussed in Section 4.3.

Figure 8 shows laminar burning velocities of $\text{NH}_3/\text{CO}/\text{air}$ flames, as a function of ϕ , measured and predicted for $x_{\text{CO}} = 0.2, 0.4, 0.6$ and 0.8 . The predicted results are lower than the experimental values of S_L for most ϕ values studied with $x_{\text{CO}} = 0.4$ and 0.6 , while better agreement can be seen for $x_{\text{CO}} = 0.2$ and 0.8 . Overall, the San Diego mechanism performs best for most cases, although it presents some large discrepancies for low values of x_{CO} . The value of ϕ with the maximum S_L also increases with the increase in x_{CO} . Compared with the $\text{NH}_3/\text{H}_2/\text{air}$ flames (cf. Fig. 6), the shift in position of the S_L peak along the equivalence ratio axis is more pronounced for the $\text{NH}_3/\text{CO}/\text{air}$ flames. It is interesting to note that this tendency is very similar to the one that occurs in flames of syngas with large percentage of CO [31].

4.1.5. $\text{NH}_3/\text{CH}_4/\text{air}$ flames

Figure 9 shows laminar burning velocities of stoichiometric $\text{NH}_3/\text{CH}_4/\text{air}$ flames, as a function of x_{CH_4} , measured and computed in the present work. The figure also includes the experimental results from the work by Okafor et al. [15], who used both linear and non-linear extrapolations to get stretch-less results from the spherical flame measurements. It can be observed an excellent agreement between the present experimental results and the literature data. Moreover, an excellent agreement is also verified between the experimental and predicted results, except for the GRI-mech, which slightly underestimates the S_L values for $0.2 < x_{\text{CH}_4} < 0.8$. Finally, the S_L values show an approximately linear dependence on x_{CH_4} , which is clearly different from the blends of NH_3 with H_2 or CO (cf. Figs. 5 and 7).

Figure 10 shows laminar burning velocities of $\text{NH}_3/\text{CH}_4/\text{air}$ flames, as a function of ϕ , measured and predicted for $x_{\text{CH}_4} = 0.2, 0.4, 0.6$ and 0.8 . Consistent with Fig. 9, larger discrepancies can generally be observed for the GRI-mech results as compared with the other two mechanisms. This, however, is not true, e.g., in the range $1.2 < \phi < 1.4$, where the GRI-mech performs better. An excellent agreement is observed between the Okafor mechanism and

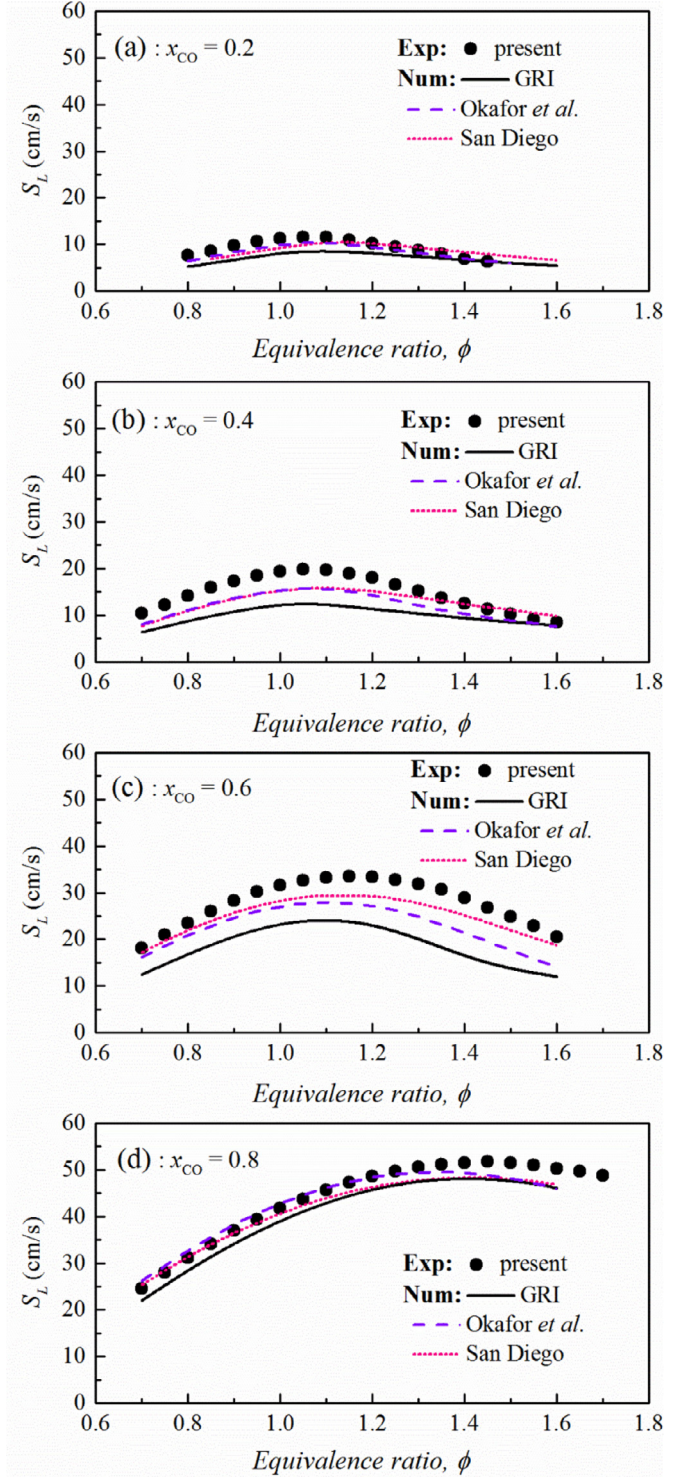


Fig. 8. Laminar burning velocities of $\text{NH}_3/\text{CO}/\text{air}$ flames, as a function of ϕ , measured in this work for $x_{\text{CO}} = 0.2, 0.4, 0.6$ and 0.8 , together with predicted values using three mechanisms.

the experimental results, particularly for $\phi < 1.2$. Moreover, the three mechanisms perform well for a wide range of ϕ at $x_{\text{CH}_4} = 0.8$, as it was the case for pure CH_4 flames (cf. Fig. 3). This suggests that CH_4 already plays the dominant role for flames where $x_{\text{CH}_4} = 0.8$. It is also interesting to note that the value of ϕ with the maximum S_L remains constant at 1.05, regardless the value of x_{CH_4} . This phenomenon is different from the blends of NH_3 with H_2 or CO, as discussed above.

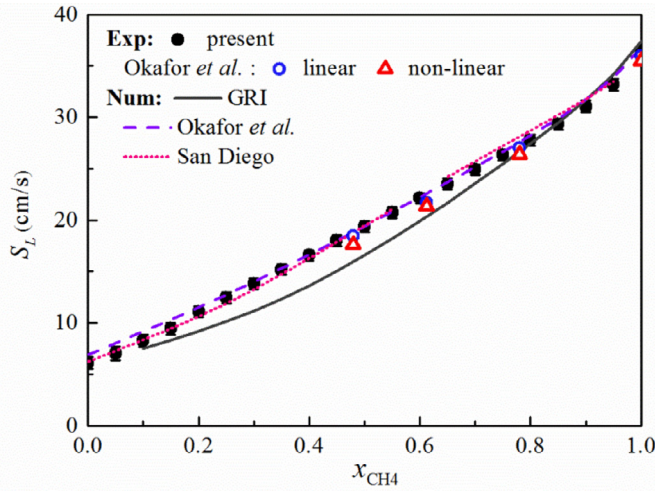


Fig. 9. Laminar burning velocities of stoichiometric NH_3/CH_4 /air flames, as a function of x_{CH_4} , measured in this work and in the work of Okafor et al. [15], and predicted laminar burning velocities using the three mechanisms.

4.2. Enhancement effect on S_L of adding H_2 , CO and CH_4 into NH_3 /air flames

To compare the degree of enhancement on S_L of adding H_2 , CO and CH_4 into NH_3 /air flames, a normalized enhancement effect parameter ξ_E is defined as:

$$\xi_E(x_M, \phi), \% = \frac{S_L(x_M, \phi) - S_{L,NH_3}(\phi)}{S_{L,NH_3}(\phi)} \quad (1)$$

where $S_{L,NH_3}(\phi)$ is the burning velocity of the NH_3 /air flames and $S_L(x_M, \phi)$ is the burning velocity of flames of NH_3 blended with M ($M=H_2$, CO and CH_4). The enhancement parameter ξ_E can be calculated using the present experimental data. According to Eq. (1), the uncertainty of ξ_E , termed as $\Delta\xi_E$, can be estimated from the uncertainties of $S_{L,NH_3}(\phi)$ and $S_L(x_M, \phi)$:

$$\begin{aligned} \Delta\xi_E &= \left| \frac{\partial \xi_E}{\partial S_L} \right| \cdot \Delta S_L + \left| \frac{\partial \xi_E}{\partial S_{L,NH_3}} \right| \cdot \Delta S_{L,NH_3} \\ &= \frac{\Delta S_L}{S_{L,NH_3}} + \frac{S_L}{S_{L,NH_3}} \cdot \frac{\Delta S_{L,NH_3}}{S_{L,NH_3}} \end{aligned} \quad (2)$$

Figure 11 shows enhancement parameter as a function of ϕ for a constant blending ratio of $x_M=0.4$. It can be seen that H_2 originates always the strongest enhancement effect on the laminar burning velocity of NH_3 /air flames, while CH_4 has the weakest enhancement effect, regardless of the value of ϕ . For $x_M=0.4$, the maximum enhancement effect is 610% for H_2 , 430% for CO and 340% for CH_4 . The degree of enhancement for H_2 , CO and CH_4 also varies with ϕ , with better enhancement effect found under fuel lean conditions (cf. Fig. 11). Under stoichiometric and fuel rich conditions, the enhancement effect remains approximately constant for the three blends. But, albeit the large error bars, a slight increase in ξ_E can be seen for H_2 and CO as ϕ increases, while a slight decrease is observed for CH_4 .

Figure 12 shows enhancement parameter ξ_E as a function of x_M for $\phi=0.85$, 1.0 and 1.25. As in Fig. 11, there is $\xi_E(H_2) > \xi_E(CO) > \xi_E(CH_4)$ for most conditions except that under lean (Fig. 12a) and stoichiometric (Fig. 12b) conditions for $x_M < 0.2$. The enhancement dependence from x_M is nearly linear for CH_4 , non-monotonic for CO and nearly exponential for H_2 . It should be noted from Fig. 12(a–c) that H_2 , CO and CH_4 have all maximum ξ_E above 400%, indicating their good potential to increase the laminar burning velocity of NH_3 /air flames. This recognition should be very important for the industrial usage of NH_3 combustion in the future.

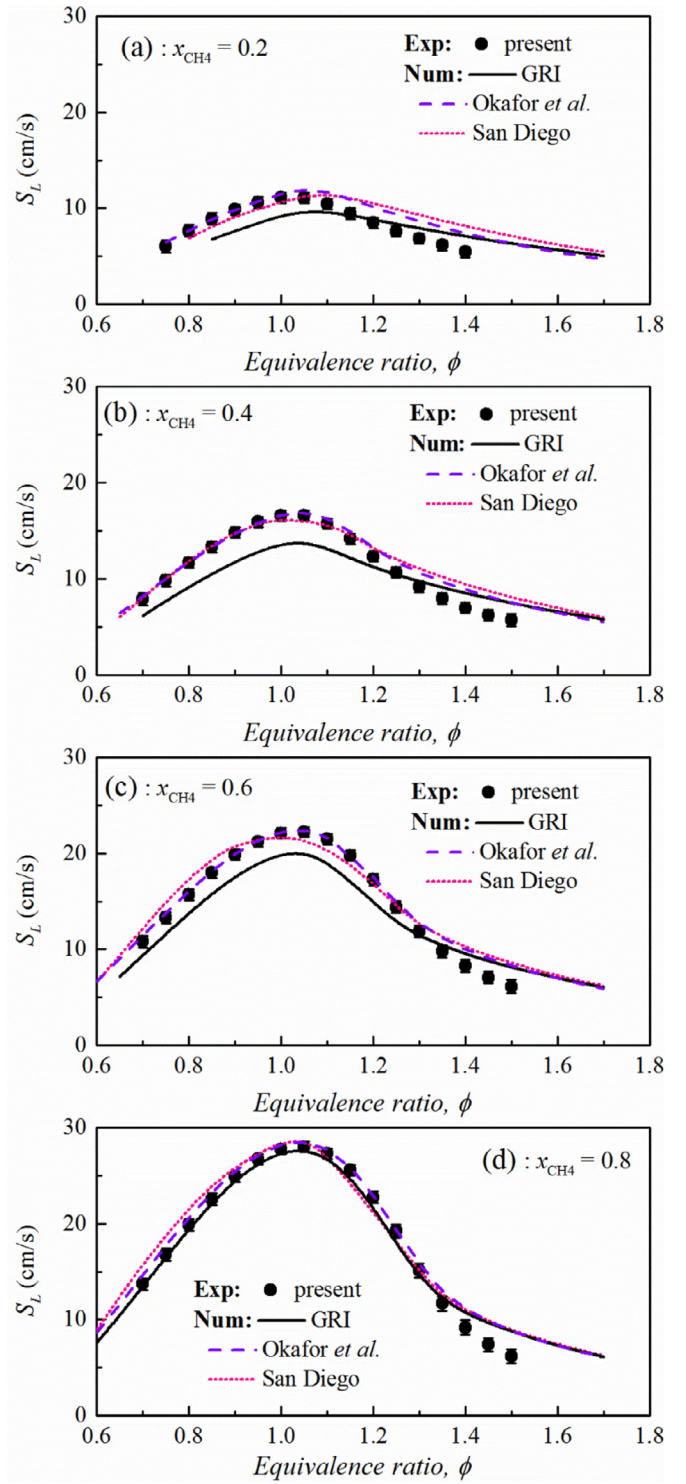


Fig. 10. Laminar burning velocities of NH_3/CH_4 /air flames, as a function of ϕ , measured in this work for $x_{CH_4}=0.2, 0.4, 0.6$ and 0.8 , together with predicted laminar burning velocities using three mechanisms.

4.3. Kinetic analyses

To investigate more deeply the reasons behind the non-monotonic relationship of S_L with CO mole fraction in Fig. 7 and those behind the large discrepancies between experimental and predicted data shown in Section 4.1, detailed kinetic analyses were carried out using the Okafor et al. [15] and the San Diego

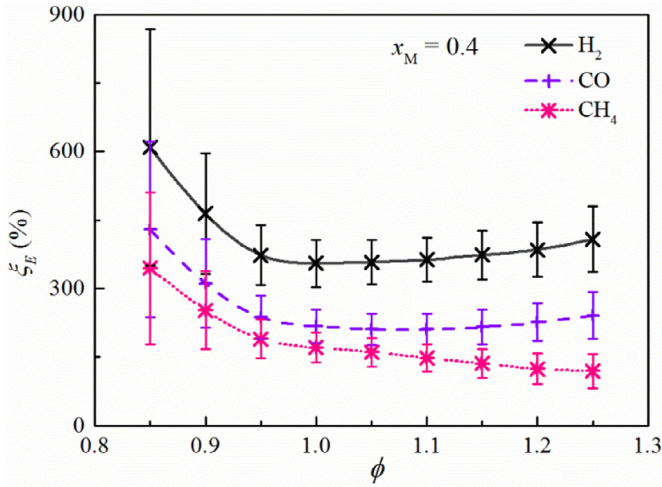


Fig. 11. Enhancement effect on laminar burning velocity of adding H₂, CO and CH₄ into NH₃/air flames for a constant blending ratio of 0.4.

[41] mechanisms, which showed better predictions than the GRI-mech 3.0 for the ammonia flames studied here.

Figure 13 shows the maximum mole fractions of the H and OH radicals for stoichiometric NH₃/CO/air flames. This figure can help to explain the non-monotonic dependence of S_L on x_{CO} observed in Fig. 7 since the H and OH radicals have a large impact on the burning velocities of NH₃ flames [15]. Figure 13 shows that pure CO/air flames are nearly free of H and OH radicals, leading to very low values of S_L , as discussed in the literature [54]. When adding small amounts of NH₃ to the CO/air mixture (cf. Fig. 13 from right to left), the H radicals generated from the NH₃ decomposition leads to the coupling of the CO oxidation and the H₂/O₂ mechanisms, with the consequent rapid increase of H and OH radical pools that lead to the rapid increase of S_L observed in Fig. 7. This situation is very alike the CO oxidation in moist air [55]. However, when the amount of NH₃ in the fuel mixture exceeds a critical value, i.e., $x_{CO}=0.85$ (cf. Fig. 7), the H and OH radical pool shrinks, leading to a decrease of S_L . It is interesting to note that the shape of the maximum H radical mole fraction line is nearly the same as that of the simulated burning velocity line, regardless of the mechanism used. This finding indicates that the impact of the H radical on the burning velocity of NH₃/CO/air flames is more important than that of the OH radical.

Figure 14 shows the correlation between the maximum H radical mole fraction and the predicted burning velocity for all cases studied in this work. The correlation is not linear because for flames with the same burning velocity for the same value of x_M , the fuel-rich side often has a larger maximum H mole fraction. This statement is exemplified in Fig. 14 for CH₄/air flames calculated using the Okafor et al. [15] mechanism, through the dash-dot (rich) and dot (lean) lines.

Based on this positive correlation shown in Fig. 14, the over- or under-prediction of S_L can be simply attributed to the over- or under-prediction of the H radical pool. While the thermodynamic and transport properties are considered to be correct, the sources of the discrepancies can result from (1) neglecting of reactions and (2) inaccuracy of rate parameters of existing reactions. From the large underestimation of the three mechanisms around $x_{CO}=0.5$ in Fig. 7, it may be straightforward to anticipate that some reactions between C and N elements containing species, represented later by C–N reactions, are neglected. However, the analyses below prove that this suspicion is wrong.

Figure 15 shows the normalized sensitivity coefficients of the burning velocity of the 10 most rate limiting reactions in

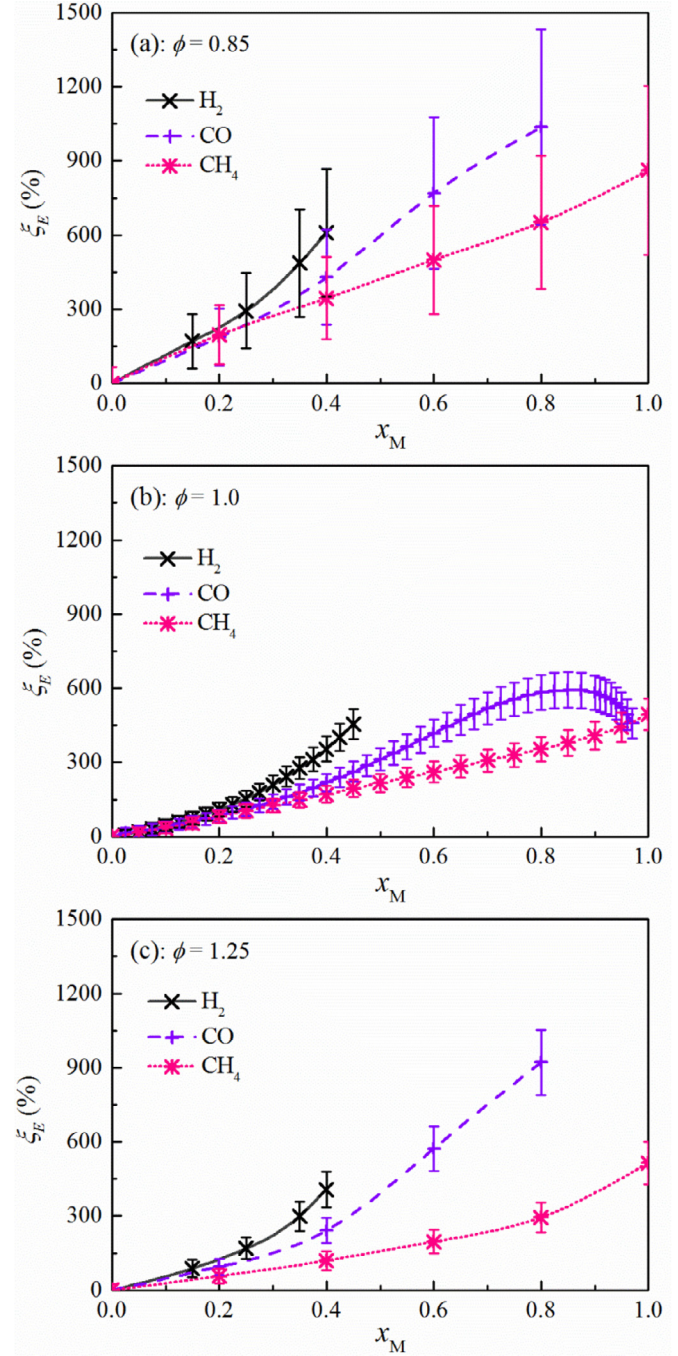


Fig. 12. Enhancement effect on laminar burning velocity of adding H₂, CO and CH₄ into NH₃/air flames at $\phi=0.85$, 1.0 and 1.25.

NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air flames for $x_M=0.4$ and $\phi=0.8$, 1.0, 1.2 and 1.6 computed using the Okafor et al. [15] and San Diego [41] mechanisms. It can be seen that the reaction $H + O_2 = O + OH$ has always the largest positive sensitivity. For the NH₃/CO/air and NH₃/CH₄/air flames, even though the San Diego mechanism [41] does not include any C–N reaction, in contrast with the Okafor et al. mechanism [15], none of this kind of C–N reactions appear in Fig. 15 as the top S_L -sensitive reactions for both mechanisms. Moreover, the S_L values predicted are rather close except a few cases (cf. Figs. 7–10). These observations indicate that the interaction between C containing species and N containing species are not critical for the prediction of the burning velocity.

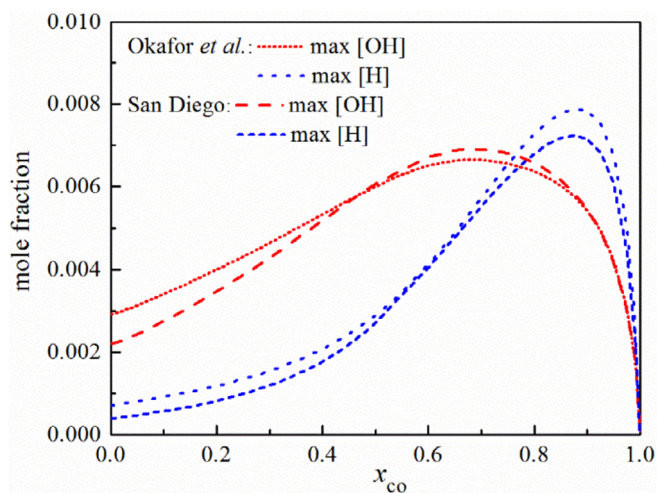


Fig. 13. Maximum mole fractions of the H and OH radicals for stoichiometric $\text{NH}_3/\text{CO}/\text{air}$ flames.

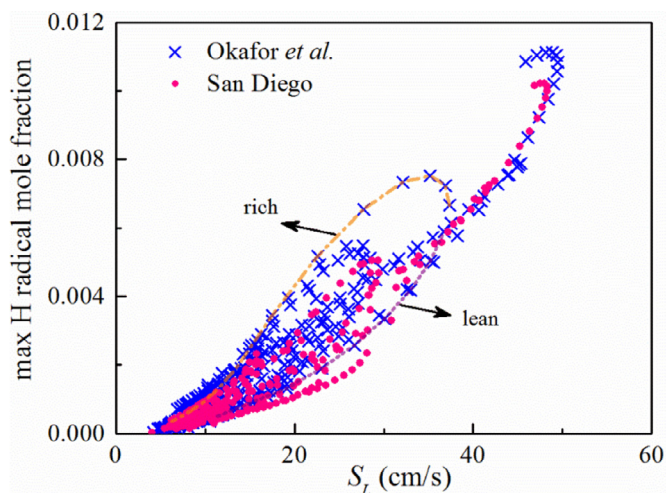


Fig. 14. Relation between the maximum mole fraction of the H radical and the predicted S_L for all cases studied in this work.

Figure 16 shows the reaction paths of the N containing species of the stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames for $x_M = 0.4$, computed using the Okafor et al. [15] and San Diego [41] mechanisms. The thickness of the arrows in Fig. 16 represents the integrated rate of species production from a chemical step or a set of chemical steps normalized by that of



in each condition. In addition, all reactions with integrated rate of species production larger than $1 \times 10^{-5} \text{ mol/cm}^2 \text{ s}$ are represented through solid lines, while the dashed lines denote some important reactions with integrated rate of species production slightly lower than $1 \times 10^{-5} \text{ mol/cm}^2 \text{ s}$. No reaction path related to C element containing species can be found in Fig. 16, indicating that C–N interactions are insignificant not only for the calculation of the burning velocity but also for the calculation of the whole flame structure.

To exclude the possibility that some C–N reactions not included in the Okafor et al. mechanism are important for the flames studied in this work, the sensitivity and reaction path analyses were also made for the mechanisms of Tian et al. [16], Mendiara and Glarborg [51], and Glarborg et al. [52], which have been used in Fig. 7. Again, it was concluded that C–N interactions were insignif-

icant. Thus the dual-fuel oxidation in $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames can be understood as parallel oxidation processes of the individual fuels, which share the same radical pools, e.g., of H and OH. This statement also agrees with the reaction route analyses of stoichiometric $\text{NH}_3/\text{CH}_4/\text{air}$ flames undertaken by Okafor et al. [15]. From this point of view, the large discrepancies observed in the predictions reported in section 4.1 are more likely due to the inaccuracy of the rate parameters of the used reactions than due to neglecting certain C–N reactions. In particular, the larger over-estimation of the experimental data made by the mechanisms of Tian et al. [16], Mendiara and Glarborg [51] and Glarborg et al. [52] (cf. Fig. 7) can be attributed to the larger cumulative error introduced by the additional reactions and species, as compared with the mechanisms of Okafor et al. [15] and San Diego [41].

Since both the Okafor et al. [15] and San Diego [41] mechanisms can predict reasonably well the burning velocities of CH_4/air flames (cf. Fig. 3), inaccuracy of the rate parameters is more likely related to the N containing reactions than to the H/O/C sub-mechanisms. Figure 16 shows that NH_3 is first dehydrated to form NH_2 , and the reactions from NH_2 to the final product N_2 can be categorized into three routes: (1) $\text{NH}_2\text{--NH--N--N}_2$; (2) $\text{NH}_2\text{--HNO--NO--N}_2\text{O--N}_2$; (3) $\text{NH}_2\text{--(N}_2\text{H}_2\text{)--NNH--N}_2$, which are ranked from high to low in terms of NH_2 consumption. The following N containing reactions show strong mechanism-dependence:



Reactions (R2) and (R3) are part of the route (3) and are included only in the Okafor et al. [15] mechanism. The omission of these two steps makes the other two NH_2 consumption routes more favored in the San Diego [41] mechanism. An investigation about the rate parameters shows that there are differences in the rate constants of the above reactions in the different mechanisms. The rate constants of reaction (R5) in San Diego [41] mechanism are always larger than in the Okafor et al. [15] mechanism, and the rate constants of reaction (R4) are slightly larger in the former when the temperature is less than 1700 K, while the rate constants of reaction (R6) are larger in the latter. As a result, the route (1) is stronger in the San Diego [41] mechanism than in the Okafor et al. [15] mechanism, and routes (2) and (3) are stronger in the Okafor et al. [15] mechanism than in San Diego [41] mechanism. Figure 15 shows that reaction (R4) is the most rate limiting N-containing reaction in rich $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames in both mechanisms, and the importance of reaction (R3) in rich $\text{NH}_3/\text{H}_2/\text{air}$ flames is proved by the Okafor et al. [15] mechanism. In light of this, the discrepancies observed in the calculation of S_L for rich flames, especially in Figs. 6 and 8, can be understood.

Figures 6, 8 and 10 reveal that the discrepancies of the predictions of the Okafor et al. [15] and San Diego [41] mechanisms under rich conditions diminish gradually on moving from Figs. 6 to 8. This tendency can be explained with the aid of Fig. 16, where the thickness of the arrows are larger for the $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames than the $\text{NH}_3/\text{CH}_4/\text{air}$ flames, indicating that the N containing reactions become less important. Since the reaction fluxes in Fig. 16 have been normalized by the reaction fluxes of the common reaction (R1) in the three fuels, the changes of arrow thickness are less likely due to the variation of S_L . Also, the C-containing reactions become more important on going

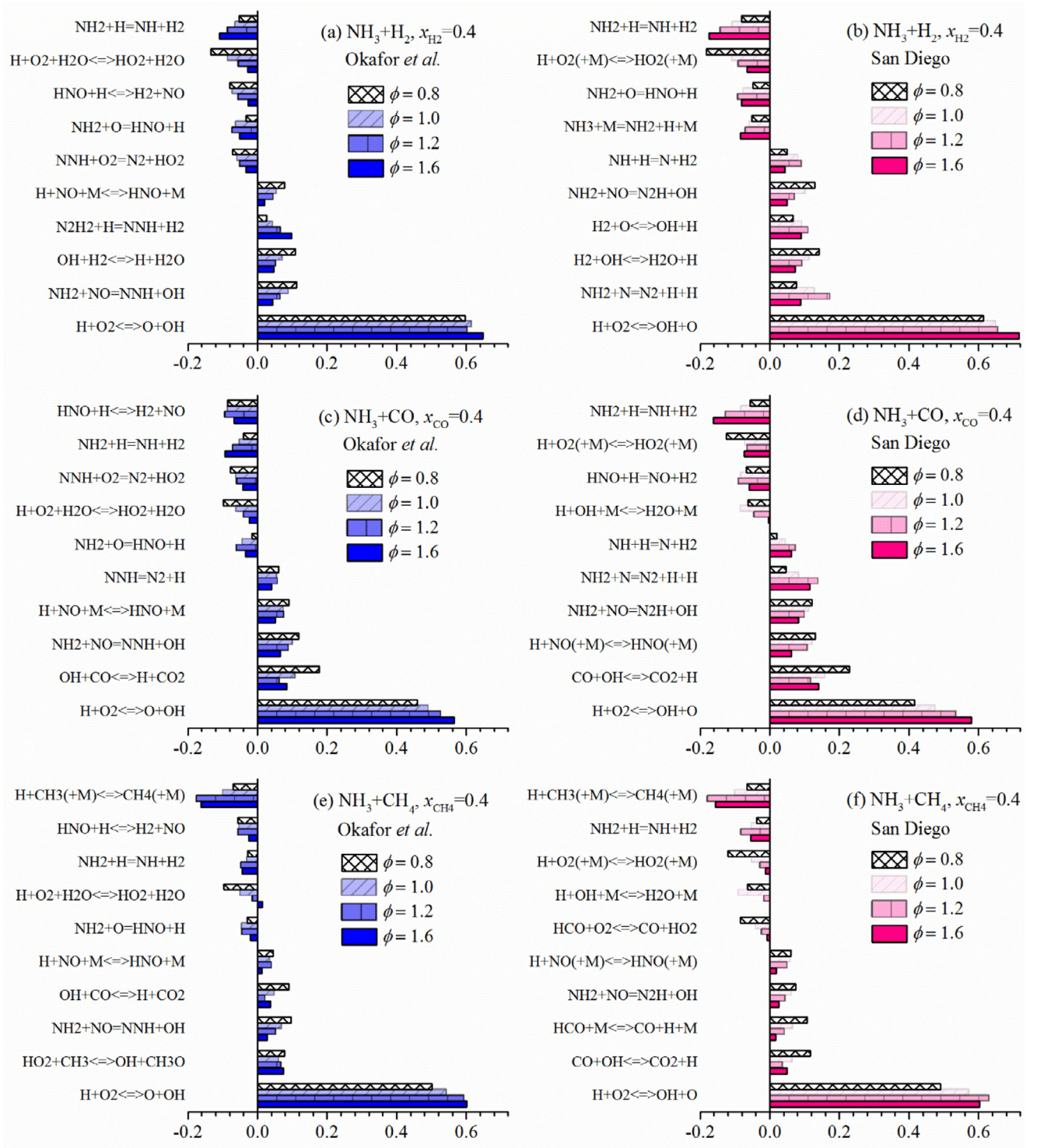


Fig. 15. Normalized sensitivity coefficients of the burning velocity of the 10 most rate limiting reactions in $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames for $x_{\text{M}} = 0.4$ and $\phi = 0.8, 1.0, 1.2$ and 1.6 computed using the Okafor *et al.* and San Diego mechanisms.

from $\text{NH}_3/\text{H}_2/\text{air}$ to $\text{NH}_3/\text{CO}/\text{air}$, and then to $\text{NH}_3/\text{CH}_4/\text{air}$ flames in Fig. 15, while the sensitivities of the most rate limiting N-containing reactions have nearly the same values, regardless of the mechanisms, e.g., R4, and



and



leading to more similar predictions for different mechanisms. Although these N-containing reactions do not affect the predictions from the different mechanisms, their inaccuracy makes the modeling results to deviate from the measured data, especially under very rich conditions (cf. Fig. 10). Figure 10 shows that the discrepancy between experimental data and predictions increases for larger values of x_{CH_4} . If this discrepancy would come from problems with the measured results, a larger disparity would exist for pure CH_4 flames in Fig. 3 because the experimental method and

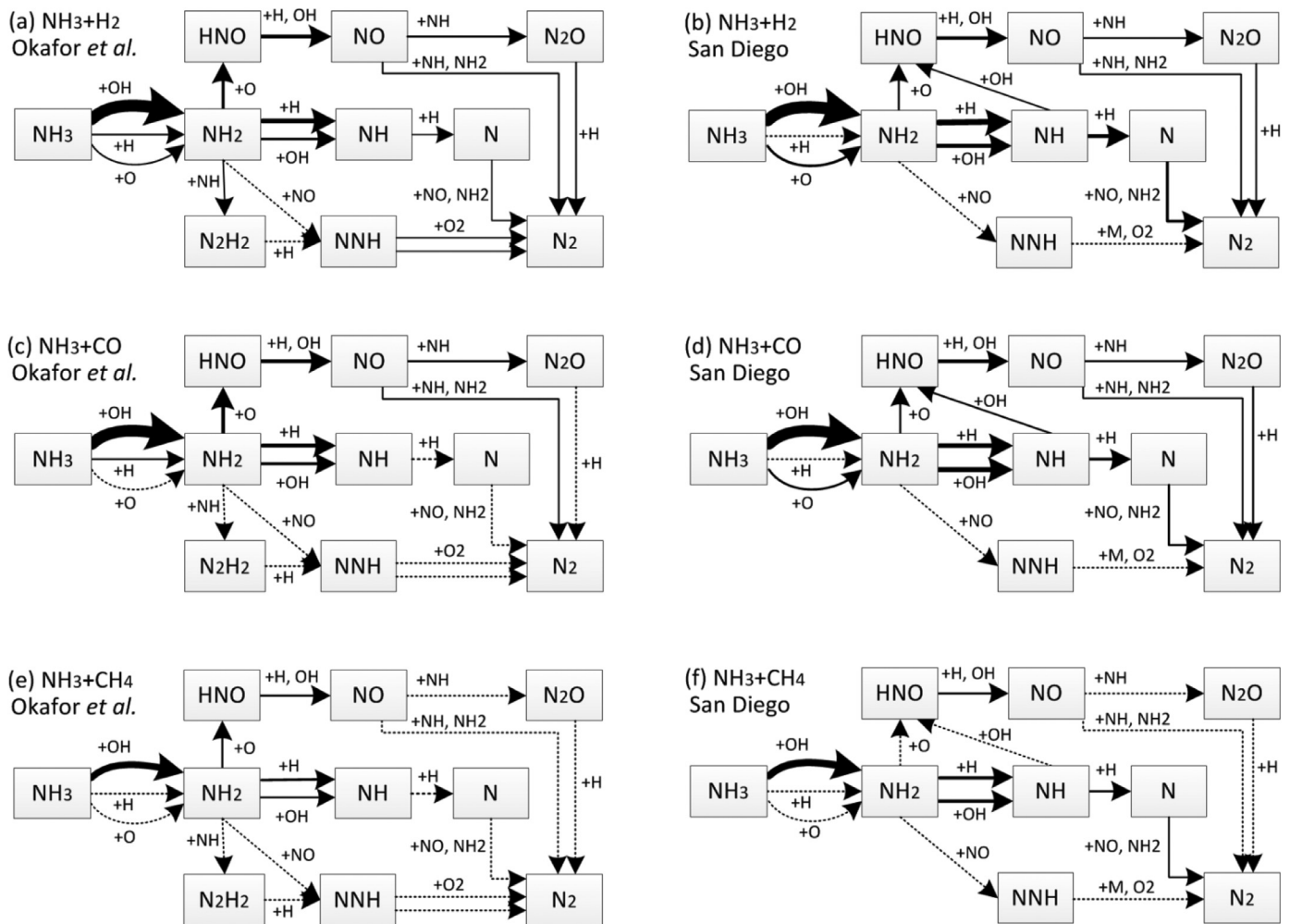


Fig. 16. Reaction paths of the N containing species for stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames for $x_M = 0.4$, computed using the Okafor et al. and San Diego mechanisms.

system used were the same. However, Fig. 3 shows that discrepancy between measurements and predictions is rather small for flames with $x_{\text{CH}_4} = 1.0$ flames, so that this possibility is invalid.

5. Conclusions

The present detailed investigation provides accurate experimental data of laminar burning velocities of NH_3 blended with H_2 , CO and CH_4 burned in air at atmospheric pressure. The experimental results and the kinetic modeling based on three literature mechanisms reveal that:

1. Blending NH_3 with H_2 is the most effective manner to increase the burning velocity of NH_3 based fuel mixtures, being the S_L values the highest ones, as compared with blending NH_3 with CO or CH_4 . In contrast, blending NH_3 with CH_4 is the less effective approach to increase the burning velocity of NH_3 based fuel mixtures.
2. The laminar burning velocity of the mixtures has different dependence on the mole fraction of the components in the fuel blend, being nearly linear for mixtures of NH_3 and CH_4 , non-linear but monotonic for mixtures of NH_3 and H_2 , and non-monotonic for mixtures of NH_3 and CO . This means that both $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames present laminar burning velocities lower than those of the H_2/air and CH_4/air flames, respectively. Blending NH_3 with CO not only shows that CO is

an effective additive to increase the laminar burning velocity of NH_3/air flames, but also shows that NH_3 is helpful to improve the laminar burning velocity of CO/air flames.

3. The value of equivalence ratio for which the laminar burning velocity is maximum depends on the mole fraction of the components in the fuel blend. For mixtures of NH_3 and H_2 , it is located around 1.05 and slightly increases with the mole fraction of H_2 in the blend. For mixtures of NH_3 and CO , the increase is more pronounced than that for mixtures of NH_3 and H_2 , while for mixtures of NH_3 and CH_4 it is independent of the mole fraction of CH_4 in the blend.
4. None of the kinetic mechanisms used in the present work can accurately predict the laminar burning velocity for most of the measured conditions. This indicates that the existing mechanisms are not especially suitable to model flames of NH_3 or those blended with H_2 , CO or CH_4 . Sensitivity and reaction path analyses indicate that the oxidation of NH_3 blended with the additive fuels considered in this work can be understood as the parallel oxidation processes of the individual fuels, which share the same radical pools of, e.g., H and OH , and the source of discrepancy between the experimental and modeling results is related to the inaccuracy of the rate parameters of the N-containing reactions. In this regard, the present detailed experimental data is of special value for the development and validation of kinetic mechanisms.

All the measured laminar burning velocities together with their uncertainties obtained in this work are summarized in the Supplementary material.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.05.003.

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